CausalPose-3D: Online Pose Estimation and Consistent Reconstruction of Unknown Objects under Degenerate Observations

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***Abstract*—Long-horizon 2D point trajectories and detector-free feature matchers provide critical data association for tracking unknown rigid objects during robotic manipulation, yet their 3D geometric constraints frequently become rank-deficient, disconnected from the initial coordinate gauge, or version-inconsistent with accumulated surfaces when sensor depth is sparse or local geometry is degenerate. Across a systematic empirical audit spanning 31 perception, matching, tracking, and reconstruction frameworks on public benchmarks (YCBInEOAT, Laval Vicon motion capture, YCB-M, DexYCB) and humanoid bimanual manipulation recordings, we demonstrate a counter-intuitive failure phenomenon: increasing pose availability or lowering internal factor-graph optimization cost can degrade strict pose accuracy and accumulated 3D surface quality. On Laval Cookiejar, repairing two upstream Point2Pose implementation defects (delayed-keyframe attribution and solver-return consistency) raises pose availability from 207 to $239/295$ frames and failure-inclusive ADD AUC from 0.574 to 0.635, but reduces $20\,\mathrm{mm}$ origin successes from 136 to 120 and drops $10\,\mathrm{mm}$ model surface coverage C_{10} from 76.82% to 61.79%. To resolve these failure modes, we present CausalPose-3D (*Causal Observability-Conditioned 6D Pose Tracking and 3D Surface Reconstruction*), a unified scientific framework built upon and extending BundleTrack and Point2Pose. We formalize unanchored $\mathrm{SE}(3)$ gauge drift ($28/33$ Cookiejar and $6/16$ Clock-full graph updates disconnected from the prior), Schur-complement local rank deficiency after nuisance landmark elimination ($8/34$ connected updates with $r_i < 6$), and unreconciled TSDF pose versions ($17/19$ fused keyframes revised without surface updates), and evaluate four controlled Point2Pose-branch interventions: prior-connected admission (G), marginal rank gating (R), immutable feature-birth observation ledgers (M), and causal surface-history rebuild transactions (S). In our primary CausalPose-3D native registration and `CausalObjectSurface` branch, we resolve CUDA RANSAC parallel-reduction non-determinism ($18.00\,\mathrm{mm}/6.22^\circ$ split converged to a single winner across 100 calls), $\mathrm{SE}(3)$ inverse-coefficient precision loss in `storeFrame` (raising Clock and Cookiejar strict successes from $135/107$ to $191/136$), bounded-window `NO_BA` non-keyframe retirement (reducing recovery and stress endpoint RSS by $928.4\,\mathrm{MiB}$ and $1{,}994.0\,\mathrm{MiB}$ with byte-identical outputs), and array-native support-constrained $1{,}500$-point surface selection (improving public C_{10} from $40.57\%/52.34\%$ to $85.10\%/85.05\%$ and task surface coverage from 71.78% to 89.97% while cutting selector latency by 33%). Finally, CausalPose-3D incorporates two bit-exact runtime execution adapters—*same-thread SAM 2 GPU-window CPU preparation overlap* and *request-scoped evidence deduplication with $O(\text{runs})$ integer RLE run-interval arithmetic, vectorized coplanar RANSAC, and request-boundary cyclic GC control*—reducing exclusive object-world wrapper overhead by 72.9% ($23.17 \rightarrow 6.28\,\mathrm{ms}$), eliminating $140.58\,\mathrm{ms}$ mid-request Generation-2 GC stalls, and lowering warm HTTP endpoint latency from $155.33/219.06\,\mathrm{ms}$ (p_{50}/p_{95}, $176.40\,\mathrm{ms}$ initial baseline) to $127.80/153.40\,\mathrm{ms}$ with zero bit divergence across 499 online equivalence checks.**

***Index Terms*—CausalPose-3D, RGB-D object tracking, 3D surface reconstruction, factor-graph observability, degenerate geometry, teleoperation assistance, bit-exact runtime optimization, reproducibility.**

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I. Introduction

ONLINE 6-DoF object pose tracking and 3D surface reconstruction from onboard RGB-D streams provide essential spatial awareness for remote humanoid teleoperation and bimanual manipulation [1]–[3]. Without pre-loaded CAD models, a manipulation perception system must initialize an object coordinate gauge from the first segmented RGB-D observation, track relative rigid motion across occlusions and depth dropouts, and accumulate a geometrically trustworthy 3D surface for operator guidance.

A common assumption in model-free tracking benchmarks is that returning a higher fraction of non-lost poses—or achieving a lower internal optimization residual—naturally yields a more reliable 3D reconstruction. Our empirical investigation refutes this assumption. Under sparse depth support, self-occlusion, or locally degenerate geometry, long-horizon 2D point trajectories and dense correspondences frequently constrain only a subspace of the joint pose-landmark state. Admitting unconstrained or unanchored updates into persistent object maps and volumetric/surfel surfaces can corrupt previously verified geometry even while increasing frame-level pose availability.

To prevent implementation artifacts from being conflated with algorithmic gains, we introduce **CausalPose-3D** (*Causal Observability-Conditioned 6D Pose Tracking and 3D Surface Reconstruction*) and strictly distinguish three methodological entities throughout this study: (1) the upstream BundleTrack algorithm [1] (pinned to `ffa67d425240`), (2) the upstream Point2Pose causal point-trajectory tracker and TSDF reconstructor [2] (pinned to `51856226610d`), and (3) our proposed **CausalPose-3D** algorithmic framework, which synthesizes observability-conditioned factor-graph and causal surface transactions (F, G, R, M, S) on the Point2Pose branch with deterministic CUDA RANSAC, exact $\mathrm{SE}(3)$ inverse-pose transfer, guarded `NO_BA` frame retirement, lineage-audited `CausalObjectSurface` (`RevisableSurfelMap`) fusion, and bit-exact runtime execution on the BundleTrack branch. All candidate estimators execute under a fail-closed

shadow protocol consuming identical registered RGB-D frames and SAM 2 video segmentation masks bound by SHA-256 receipts, without overwriting live teleoperation wire authority.

Our contributions are fourfold:

1) **Failure Mechanism Formulation and Observability Audit (Sections III–IV):** We formalize three coupled failure modes in long-horizon RGB-D tracking and fusion: (i) temporal attribution and solver-return inconsistency (Fork F), (ii) unanchored $\mathrm{SE}(3)$ gauge drift and nuisance-coupled rank deficiency after orthogonal left-null-space landmark elimination $\bar{\mathbf{A}}_x = (\mathbf{I} - \mathbf{A}_y \mathbf{A}_y^{+})\mathbf{A}_x$, and (iii) pose-version desynchronization between factor-graph revisions and accumulated surfaces. We evaluate controlled interventions (G, R, M, S) across 704 Vicon-referenced frames and 71 exact native factor-graph replays.
2) **Native BundleTrack Determinism, Precision, and Memory Governance (Section IV-B):** We identify and fix parallel-reduction tie-breaking races and count-comparison directions in native CUDA RANSAC, repair float-to-matrix inverse-coefficient truncation in `CUDACache::storeFrame`, and introduce guarded `NO_BA` non-keyframe retirement that reclaims 928.4–1,994.0 MiB of host RSS under tracking loss while preserving 100% bit-exact pose and surface outputs.
3) **Lineage-Audited Surface Fusion and Array-Native Budget Selection (Section IV-C):** By tracing 1,512,357 surfel contributions to 1,960,512 source pixels and continuous mesh rays, we separate pose-induced surface tails from raw boundary measurements, and implement an array-native `support10/coarse12mm` 1,500-point selector on `RevisableSurfelMap` that cuts selection latency by 33% while raising model coverage C_{10} from 40.57%/52.34% to 85.10%/85.05%.
4) **31-Framework Context and Bit-Exact Runtime Execution Optimization (Sections IV-D–V):** Within our benchmark of 31 perception, depth, pointmap, matching, and tracking frameworks (including EfficientLoFTR, XFeat, and ALIKED+LightGlue shadow trials), we introduce `SAM2GpuWindowOverlapScope` and `RequestEvidenceDeduplicationScope` (O(runs) exact integer RLE arithmetic, cross-stage surface/medoid reuse, BLAS-screened coplanar RANSAC, and request-boundary cyclic GC control), cutting warm HTTP endpoint latency by 27.6% (176.40 → 127.80 ms p_{50}, 219.06 → 153.40 ms p_{95}) with 100% bit-exact parity.

II. Related Work and Architectural Landscape

Model-free RGB-D trackers estimate camera-local object motion from an initial segmented frame. BundleTrack [1] combines pairwise feature matching with keyframe pose-graph optimization over sparse 3D correspondences and dense point-to-plane residuals. BundleSDF [3] jointly optimizes a keyframe pose graph and a neural implicit signed distance field, whereas FoundationPose [4] unifies CAD-based and few-view reference-assisted pose estimation and tracking. Point2Pose [2] introduces long-horizon 2D point tracking (Online BootsTAPIR [8]) across occlusions, lifting 2D tracks to 3D observations for multi-hypothesis SVD-RANSAC registration, GTSAM [7] Levenberg-Marquardt pose-landmark graph optimization, and incremental TSDF fusion. While Point2Pose's point identity persistence, multi-object query management, and graph/TSDF architecture are established upstream contributions, we show that its unguarded graph commits and decoupled TSDF integration can degrade surface fidelity when 2D tracks lack well-distributed 3D depth support. In parallel, replacing detector-free LoFTR [9] with faster matchers such as EfficientLoFTR [10], semi-dense XFeat [11], or sparse ALIKED+LightGlue [12], [13] exposes a strict accuracy–latency frontier when foreground masks contain $< 1\%$ valid sensor depth.

State estimation under geometric degeneracy has been studied in point-cloud registration and SLAM by Zhang *et al.* [14] and Tuna *et al.* (X-ICP [15]), which project optimization increments onto well-conditioned Hessian subspaces. However, evaluating pose conditioning with landmarks held fixed overestimates available information when the object map itself is uncertain. In dense mapping, BundleFusion [16] de-integrates and re-integrates RGB-D frames when bundle adjustment revises camera poses. For unknown dynamic objects, however, re-integrating historical keyframes with revised graph poses only improves the surface if the graph revisions themselves are metrically accurate and anchored to the initial gauge.

III. Problem Formulation and Failure Mechanisms under Degenerate Observations

A. Observation Provenance, Coordinate Gauge, and Metrics

Let each tracking observation tuple be bound as

$$\mathcal{O}_t = \big(\text{obj_id}, f_t, \{(u_j, v_j, d_j, \text{id}_j, \text{valid}_j)\}_{j=1}^{M_t}, \mathbf{K}, M_t^{\text{seg}}\big), \tag{1}$$

distinguishing sensor-measured 3D points $\mathbf{z}_{t,j} = \pi^{-1}(u_j, v_j, d_j; \mathbf{K})$ from depth-missing 2D tracks, RANSAC proposals, numerical initializations, and the fixed Frame-0 gauge O_{obj_0}. Each tracker's initial gauge is aligned once using the first reference pose, with Frame 0 excluded from score denominators. For post-initial frames, ADD measures mean corresponding model-vertex distance and ADD-S measures mean nearest-vertex distance; unavailable outputs receive zero recall at all thresholds in 0–100 mm normalized recall AUC.

For an accumulated reconstructed point set S and reference surface point set M evaluated on a common 4 mm grid without ICP drift masking, we simultaneously report directional precision P_ϵ and reference coverage C_ϵ:

$$P_\epsilon = |\{s \in S : \min_{m \in M} \|s - m\|_2 \le \epsilon\}|/|S|, \tag{2}$$

$$C_\epsilon = |\{m \in M : \min_{s \in S} \|m - s\|_2 \le \epsilon\}|/|M|, \tag{3}$$

alongside directional p_{95} distances F_{95} ($S \to M$) and R_{95} ($M \to S$). Deleting uncertain points can inflate P_ϵ while collapsing C_ϵ; a valid surface improvement must account for both directions.

B. Mechanism 1: Temporal Attribution and Solver-Return Inconsistency

Source inspection of pinned Point2Pose (`51856226610d`) reveals two internal state-contract defects:

1) *Historical Keyframe Attribution Overwrite:* When a pending keyframe from historical frame $f_{\text{kf}} < f_t$ is promoted after multi-frame verification, `ModularPipeline.step` writes the graph-optimized keyframe pose $\mathbf{T}_{\text{kf}}$ into current `obj.pose` without checking $f_{\text{kf}} == f_t$.
2) *Refined-Pose Return Mismatch:* In `SVDClusterRANSACRegister.register`, when local refinement updates a hypothesis, the function recomputes final inliers and residuals using refined $\mathbf{T}_{\text{ref}}$ but returns pre-refinement $\mathbf{T}_{\text{raw}}$.

C. Mechanism 2: Unanchored Gauge Drift and Nuisance-Coupled Rank Deficiency

In GTSAM [7], numerical `Values` provide linearization points, whereas only `Factors` constrain the state. Native Point2Pose places a single prior factor on initial keyframe pose $\mathbf{X}_0 \in \mathrm{SE}(3)$ and connects subsequent poses $\mathbf{X}_m$ ($m \in \mathcal{V}_p$) and 3D landmarks $\mathbf{p}_n \in \mathbb{R}^3$ ($n \in \mathcal{V}_\ell$) via `BearingRangeFactors` (from registration inliers) and conditional `BetweenFactors` (admitted only when all-correspondence mean residual $< 0.01\,\mathrm{m}$).

Unanchored Gauge Freedom: Let $C_0 \subseteq \mathcal{V}_p \cup \mathcal{V}_\ell$ denote the factor-graph connected component containing the sole initial prior $\mathbf{X}_0$. For any disconnected component $C' \cap C_0 = \emptyset$, transforming all inverse poses and landmarks in C' by an arbitrary global rigid transformation $\mathbf{Q} \in \mathrm{SE}(3)$, $\mathbf{X}'_m = \mathbf{Q}\mathbf{X}_m$ and $\mathbf{p}'_n = \mathbf{Q}\mathbf{p}_n$, leaves every relative observation $(\mathbf{X}'_m)^{-1}\mathbf{p}'_n = \mathbf{X}_m^{-1}\mathbf{p}_n$ and relative-pose factor invariant. Thus Levenberg-Marquardt damping can converge numerically with non-increasing cost while C' drifts freely relative to O_{obj_0} and overwrites persistent frontend map coordinates.

Nuisance-Eliminated Local Observability: Even when $\mathbf{X}_i \in C_0$, topological connectivity does not imply 6-DoF observability. In the whitened, prior-conditioned local linear system around the current linearization point, let $\delta\mathbf{x} \in \mathbb{R}^{6|\mathcal{V}_p|}$ denote pose tangent increments and $\delta\mathbf{y} \in \mathbb{R}^{3|\mathcal{V}_\ell|}$ denote landmark increments:

$$\min_{\delta\mathbf{x},\delta\mathbf{y}} \|\mathbf{A}_x\delta\mathbf{x} + \mathbf{A}_y\delta\mathbf{y} - \mathbf{b}\|_2^2. \tag{4}$$

Eliminating uncertain landmarks via orthogonal projection onto the left null space of $\mathbf{A}_y$ yields the reduced pose Jacobian $\bar{\mathbf{A}}_x = (\mathbf{I} - \mathbf{A}_y\mathbf{A}_y^+)\mathbf{A}_x$ and marginal pose information matrix:

$$\bar{\mathbf{H}}_x = \bar{\mathbf{A}}_x^\top\bar{\mathbf{A}}_x = \mathbf{A}_x^\top(\mathbf{I} - \mathbf{A}_y\mathbf{A}_y^+)\mathbf{A}_x \preceq \mathbf{A}_x^\top\mathbf{A}_x. \tag{5}$$

Treating uncertain landmarks as known ($\mathbf{A}_x^\top\mathbf{A}_x$) strictly overestimates pose conditioning. The locally constrained dimension of current pose i is given by the column-deletion rank difference:

$$r_i = \mathrm{rank}(\bar{\mathbf{A}}_x) - \mathrm{rank}(\bar{\mathbf{A}}_{x,-i}) \in \{0,\dots,6\}, \tag{6}$$

where $\bar{\mathbf{A}}_{x,-i}$ removes the six columns of pose i. Under non-collinear 3D correspondences anchored to C_0, one shared landmark supplies $r_i = 3$, two supply $r_i = 5$, and three non-collinear landmarks supply $r_i = 6$.

D. Mechanism 3: Pose-Version Desynchronization in Surface Fusion

When `KeyFrameGraph.update` revises a historical keyframe pose from its initial integration estimate $\mathbf{T}_\tau^{(\tau)}$ to $\mathbf{T}_\tau^{(t)}$ ($t > \tau$), `ModularPipeline.step` updates the sparse landmark map, but `SDFBuilder.integrate_keyframe` only integrates the newly promoted keyframe into the TSDF volume without de-integrating or re-projecting earlier observations fused at stale poses $\{\mathbf{T}_\tau^{(\tau)}\}$. Consequently, the sparse tracker map and dense TSDF surface diverge in coordinate version history.

IV. The CausalPose-3D Framework: Constrained State Updates, Native Governance, and Zero-Loss Execution

A. Controlled Graph and Surface Interventions (F, G, R, M, S)

Within **CausalPose-3D**, we first isolate five modular, default-off interventions on the Point2Pose causal trajectory architecture:

- **Correctness Baseline (F):** Enforces frame-scoped pose attribution ($f_{\text{kf}} == f_t$ before updating `obj.pose`) and returns refined pose $\mathbf{T}_{\text{ref}}$ consistently with its inliers and residuals.
- **Prior-Connected Graph Admission (G):** Admits optimized keyframe and landmark updates into the frontend map only for variables residing in the connected component C_0 of initial pose prior $\mathbf{X}_0$. Disconnected components retain their optimizer state for potential later reconnection without corrupting the active map.
- **Marginal Rank Gating (R):** Extends G by computing r_i in (6) across three numerical tolerances $(\{0.1\times, 1\times, 10\times\}\epsilon\max(\mathrm{shape}(\bar{\mathbf{A}}_x))\sigma_{\max}(\bar{\mathbf{A}}_x))$ and withholding graph commit when current pose $r_i < 6$.
- **Feature-Birth Observation Ledger (M):** Records immutable original camera-space 3D measurements $(\text{obj_id}, f_{\text{birth}}, f_{\text{admit}}, \text{track_id}, \mathbf{z}_{\text{cam}})$ when features are first sampled, and inserts deduplicated native `BearingRangeFactors` at the birth pose once registration instantiates the landmark.
- **Causal Surface-History Transaction (S):** Maintains an immutable ledger of integrated depth/RGB observations and their pose versions. Whenever an admitted graph update modifies a previously integrated keyframe pose, S deterministically rebuilds a temporary volume from causal history using current pose versions $\mathbf{T}_\tau^{(t)}$ before integrating new evidence, committing atomically upon grid and identity verification.

B. BundleTrack Native Determinism, Inverse Precision, and Frame Retirement

Auditing our primary BundleTrack C++/CUDA pose owner and session adapter uncovered four concrete engineering improvements:

1) **`NO_BA` Status Integrity:** A two-line adapter guard prevents retained native `NO_BA` states from being misclassified as historical candidates on cache misses, eliminating 6 false historical promotions while preserving all $34/38$ valid task candidates.
2) **CUDA RANSAC Deterministic Winner Selection:** In `BundleTrack/src/cuda/`, two identical 13-correspondence CPU PCL inputs produced divergent 6-inlier subsets differing by $18.00\,\mathrm{mm}$ and 6.22° due to parallel atomic tie-breaking and reversed count-comparison inequalities. Applying the `stable-winner` and count-direction repair collapses 100 repeated native RANSAC+Procrustes calls onto a single deterministic output.
3) $\mathrm{SE}(3)$ **Inverse-Coefficient Precision in `storeFrame`:** Replacing lossy bulk cast transfer with exact element-wise Eigen-to-`float4x4` inverse-pose coefficient copying in `CUDACache::storeFrame` ensures pinhole backprojection accuracy $\leq 1.30 \times 10^{-7}\,\mathrm{m}$.
4) **Guarded `NO_BA` Non-Keyframe Retirement:** Under continuous tracking loss, `Bundler` previously retained all blocked frames in GPU/host memory. We invoke `Bundler::forgetFrame` on non-keyframes beyond the window_size $+\,1$ budget only when the most recent window_size $+\,1$ frame IDs are contiguous and all carry status `NO_BA`, while protecting first, current, neighbor, local, keyframe, and pose-cached frames.

C. Array-Native Support-Constrained 1,500*-Point Surface Selection*

To transmit compact object surfaces within the 1,500-point wire budget without sacrificing geometric coverage, we extend `RevisableSurfelMap.snapshot` with an array-native `support10/coarse12mm` selector (`owner-array-snapshot-shadow-20261101`). Rather than materializing full Python dictionary lists before truncation, the selector operates directly on contiguous NumPy coordinate and support arrays, prioritizing voxels with observation support ≥ 10 across $12\,\mathrm{mm}$ coarse spatial cells before backfilling remaining capacity in deterministic encounter order.

D. Zero-Loss Runtime Execution: GPU-Window Overlap and $O(\textit{runs})$ *Evidence Deduplication*

Critical-path profiling of the synchronous `/world_model/observe` endpoint revealed that sequential SAM 2 and native BundleTrack execution ($95.87/118.98\,\mathrm{ms}$ median/p_{95}) was compounded by $\sim 48\,\mathrm{ms}$ of redundant host CPU work and mid-request cyclic GC stalls. We introduce two stacked, bit-exact perception adapters:

1) Same-Thread SAM 2 GPU-Window CPU Preparation Overlap (`SAM2GpuWindowOverlapScope`): On warm distorted frames, `SAM2VideoMemoryTracker.track` enqueues CUDA kernels for `propagate_in_video` in $\sim 9.0\,\mathrm{ms}$ and blocks the host thread for $\sim 24.2\,\mathrm{ms}$ at `(logits > 0.0).cpu().numpy()` waiting for GPU `memory_attention`. Instead of spawning background `ThreadPoolExecutor` workers that contend for the Python GIL, our adapter performs a $0.25\,\mathrm{ms}$ fail-closed pre-validation, launches `propagate_in_video`, and executes mask-independent CPU preparation *on the same thread inside the* $24.2\,\mathrm{ms}$ *GPU wait window* before calling `.cpu().numpy()`: (i) bilinear/nearest undistortion `rectify_registered_rgbd`, (ii) dual SHA-256 evidence-hash prefixes covering `header+color_rgb` and `header+color_rgb+depth_mm` ($4.61/5.53\,\mathrm{MB} = 83.3\%$ of input bytes), and (iii) private `color_bgr` and `depth_m` buffers for `PinnedBundleTrackSession`. After GPU mask readback, it reuses in-scope `sensor_mask` RLE/SHA-256 and finalizes the remaining 16.7% hash tail synchronously in $0.374\,\mathrm{ms}$, eliminating all 38 per-frame background hash threads.

2) Request-Scoped Evidence Deduplication and $O(\textbf{runs})$ Run-Interval Arithmetic (`RequestEvidenceDeduplicationScope`):

- *Exact $O(\textit{runs})$ RLE Bounding Box and Centroid:* `FoundationMaskCacheDetector` previously iterated pixel-by-pixel in Python over 80,696 foreground pixels per mask, allocating $>$ 161,000 Python `int` objects per frame ($10.38\,\mathrm{ms}$). Given RLE runs (s_k, L_k) with $e_k = s_k + L_k - 1$ on a $H \times W$ grid, let $(y_0, x_0) = \mathrm{divmod}(s_k, W)$ and $(y_1, x_1) = \mathrm{divmod}(e_k, W)$. Using exact Python integers, the row sum $\sum y$ across a multi-row run ($y_0 < y_1$) is evaluated in $O(1)$ closed form:

$$\Sigma_{y,k} = (W - x_0)y_0 + W\tfrac{(y_1 - y_0 - 1)(y_0 + y_1)}{2} + (x_1 + 1)y_1, \tag{7}$$

  and the column sum follows from linear index sums $\Sigma_{\mathrm{idx},k} = L_k s_k + L_k(L_k - 1)/2$ via $\sum x = \sum_k \Sigma_{\mathrm{idx},k} - W \sum_k \Sigma_{y,k}$, reducing box/centroid evaluation from $10.38\,\mathrm{ms}$ to $0.015\,\mathrm{ms}$ (99.85% reduction) with bit-for-bit identity.
- *Cross-Stage Surface and Medoid Reuse:* Within a request scope, we memoize `observation_digest`, `decode_compressed_depth`, and `decode_mask_rle`, and reuse Z-buffered `RegisteredRgbdFrame.select_surface(mask)` and `select_mask_centroid_depth_component` across `table_contexts`, `detector._position_detections_for_observation`, and `run_live._live_surface_projection_from_detec` ($41/41$ hits, 0 fallbacks).
- *Vectorized Coplanar AprilTag Plane RANSAC:* `fast_fit_shared_plane` caches deterministic `default_rng(0)` 240×3 index triplets by point count, screens candidate normals via BLAS matrix multiplication, and verifies top contenders in exact `float64`, reducing plane fitting on tag-visible frames from $4.47\,\mathrm{ms}$ to $0.89\,\mathrm{ms}$.
- *Request-Boundary Cyclic GC Management:* Calling `gc.collect()` after model initialization reclaims $\sim 745\,\mathrm{MB}$ of unreferenced tensors (reducing CUDA allocated memory from $813\,\mathrm{MB}$ to $67.1\,\mathrm{MB}$) and defers cyclic GC to request boundaries, eliminating 136–141 ms mid-request Generation-2 scans.

TABLE I
SELECTED REFERENCE-MASK YCBINEOAT DEVELOPMENT RESULTS. COUNTS ARE ADD-S SUCCESSES WITHIN 10 MM; AUC IS FAILURE-INCLUSIVE OVER 0–100 MM. THE LAST ACTION WAS SELECTED SUBSEQUENTLY. P2P: POINT2POSE; BT: BUNDLETRACK.

Sequence	n	P2P 10	BT 10	P2P AUC	BT AUC
Cracker	374	365	373	0.961	0.952
Mustard	736	726	736	0.946	0.951
Bleach	662	155	301	0.798	0.851
Sugar	1001	617	794	0.915	0.920
Tomato	1307	1301	1303	0.957	0.959
Bleach-hard*	440	416	418	0.948	0.930

TABLE II
OPENED VICON-REFERENCE DEVELOPMENT SEQUENCES. COUNTS USE ALL POST-INITIALIZATION FRAMES; AUC COUNTS UNAVAILABLE POSES AS ZERO. B: BUNDLETRACK; P: ORIGINAL POINT2POSE; F: CUMULATIVE CORRECTNESS FORK. THESE ARE NOT HELD-OUT RESULTS.

Sequence	Method	Avail.	$\leq$10 mm	$\leq$20 mm	$\leq 10^\circ$	AUC
Clock-hard	B	202/202	119/202	199/202	162/202	0.887
Clock-hard	P	43/202	25/202	40/202	28/202	0.188
Clock-hard	F	157/202	27/202	137/202	122/202	0.658
Clock-full	B	197/204	150/204	197/204	197/204	0.885
Clock-full	P	112/204	64/204	101/204	60/204	0.458
Clock-full	F	169/204	156/204	169/204	85/204	0.721
Cookiejar	B	295/295	109/295	214/295	112/295	0.723
Cookiejar	P	207/295	54/295	136/295	140/295	0.574
Cookiejar	F	239/295	64/295	120/295	142/295	0.635

V. EXPERIMENTAL EVALUATION AND MECHANISM ABLATIONS

A. Metric-Dependent Pose Baselines and the Availability–Surface Paradox

Table I compares Point2Pose and BundleTrack on six reference-mask YCBInEOAT [5] sequences with identical 3×3 eroded masks. BundleTrack passes more frames at 10 mm ADD-S on all six sequences, whereas Point2Pose achieves higher 0–100 mm recall AUC on Cracker and Bleach-hard. On Sugar, switching from nearest-vertex ADD-S (617 vs. 794) to corresponding-vertex ADD (206 vs. 174) reverses the winner, showing that single-threshold rankings depend on symmetry and metric conventions.

On the three Vicon-referenced Laval actions [6] sharing identical SAM 2 masks and RGB-D entries (203/205/296 frames), factorial decomposition on Clock-hard confirms that frame attribution alone yields 43/202 available poses (AUC 0.1923), solver-return consistency alone yields 158/202 (AUC 0.6260), and combined fork F yields 157/202 (AUC 0.6583) vs. original P at 43/202 (AUC 0.1881). Yet Table II and Fig. 1 expose the central paradox on Cookiejar: while F increases availability from 207 to 239/295 and ADD AUC from 0.574 to 0.635, its 20 mm origin successes drop from 136 to 120 (22 regressions vs. 9 gains on 192 mutually tracked frames, and 0/47 newly available frames within 10 mm). Simultaneously, at the fixed Frame-150 surface snapshot, F drops 10 mm model coverage C_{10} from 76.82% to 61.79% ($P_{10} = 47.48\%$), compared with $P_{10}/C_{10} = 64.12\%/99.71\%$ for BundleTrack's causal surfel map.

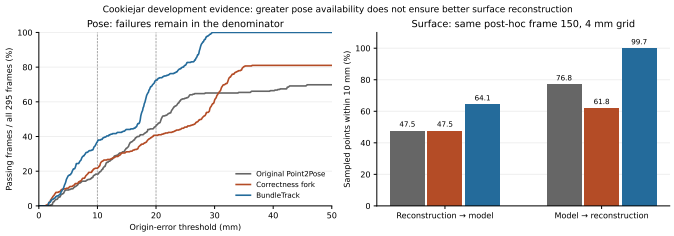


Fig. 1. **The availability–surface paradox on Laval Cookiejar.** Repair fork F increases pose availability (207 → 239/295) and ADD AUC (0.574 → 0.635), but degrades strict 20 mm pose successes (136 → 120) and reduces fixed-snapshot 10 mm surface coverage C_{10} from 76.82% to 61.79%.

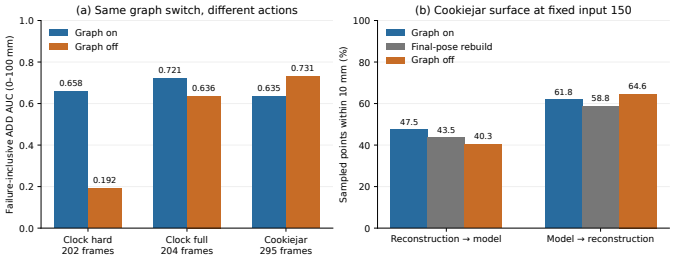


Fig. 2. **Opposing cross-action effects of disabling the global keyframe factor graph.** Turning off the graph improves Cookiejar availability/AUC (239/0.635 → 260/0.731) but severely degrades Clock-hard (157/0.658 → 43/0.192) and Clock-full (169/0.721 → 142/0.637).

B. Factor-Graph Observability and Causal Surface Transactions

Disabling the global factor graph improves Cookiejar AUC to 0.7308 but collapses Clock-hard AUC from 0.6583 to 0.1922 (Fig. 2). Passive graph auditing across all 704 records explains why: the current pose lacks a factor path to initial prior $\mathbf{X}_0$ in 28/33 Cookiejar updates and 6/16 Clock-full updates (0/22 on Clock-hard), causing 6,519 and 235 unanchored coordinate writes to active frontend map points despite non-increasing optimizer cost. Exact replay of all 71 native optimizations reveals that 8 of 34 connected non-initial updates are locally rank-deficient ($r_i < 6$) after eliminating landmark increments via (6).

Table III evaluates graph interventions G, R, and M. Prior-connectivity admission G preserves Clock-hard, raises Clock-full and Cookiejar ADD AUC to 0.739 and 0.668, and partially recovers Cookiejar surface coverage C_{10} from 61.79% to 69.94% ($P_{10} = 50.16\%$). Marginal rank gating R boosts Cookiejar 10 mm origin successes from 62 to 97 (AUC 0.718) but over-rejects partially constrained updates on Clock-hard (AUC 0.545). Adding 394/25/83 missing feature-birth factors in M makes every graph update topologically connected (0 unanchored poses), yet standalone M degrades Clock-hard/full trajectories because raw birth depth noise enters the graph without surface version reconciliation.

In the F baseline, 17 of 19 integrated Cookiejar keyframes receive later graph pose revisions while native TSDF arrays remain static; a retrospective final-pose rebuild yields $P_{10}/C_{10} = 43.49\%/58.76\%$, proving that pose-version consistency cannot rescue inaccurate graph poses. Table IV and Fig. 3 present the 2×2 causal factorial combining G and M

TABLE III
GRAPH INTERVENTIONS ON OPENED DEVELOPMENT ACTIONS. F: CORRECTNESS FORK; G: PRIOR-CONNECTIVITY ADMISSION; R: G PLUS CURRENT-POSE RANK ADMISSION; M: G PLUS MEASURED BIRTH OBSERVATIONS. ALL COUNTS EXCLUDE INITIALIZATION; REGRESSIONS ARE RETAINED.

Action	Arm	Avail.	≤10 mm	≤20 mm	≤ 10°	AUC
Clock-hard	F	157/202	27/202	137/202	122/202	0.658
Clock-hard	G	157/202	27/202	137/202	122/202	0.658
Clock-hard	R	135/202	28/202	97/202	77/202	0.545
Clock-hard	M	53/202	26/202	50/202	47/202	0.236
Clock-full	F	169/204	156/204	169/204	85/204	0.721
Clock-full	G	170/204	152/204	169/204	105/204	0.739
Clock-full	R	169/204	143/204	168/204	73/204	0.709
Clock-full	M	143/204	102/204	134/204	111/204	0.609
Cookiejar	F	239/295	64/295	120/295	142/295	0.635
Cookiejar	G	250/295	62/295	127/295	133/295	0.668
Cookiejar	R	250/295	97/295	182/295	168/295	0.718
Cookiejar	M	237/295	71/295	140/295	174/295	0.664

TABLE IV
CAUSAL SURFACE FACTORIAL ON THE SAME OPENED ACTIONS. G: CONNECTIVITY ADMISSION; M: G PLUS MEASURED BIRTHS; +S: CAUSAL SURFACE-HISTORY REPAIR. ALL COUNTS EXCLUDE INITIALIZATION.

Action	Arm	Avail.	≤10 mm	≤20 mm	≤ 10°	AUC
Clock-hard	G	157/202	27/202	137/202	122/202	0.658
Clock-hard	G+S	136/202	30/202	103/202	68/202	0.552
Clock-hard	M	53/202	26/202	50/202	47/202	0.236
Clock-hard	M+S	155/202	26/202	44/202	42/202	0.583
Clock-full	G	170/204	152/204	169/204	105/204	0.739
Clock-full	G+S	146/204	86/204	132/204	113/204	0.620
Clock-full	M	143/204	102/204	134/204	111/204	0.609
Clock-full	M+S	163/204	80/204	158/204	101/204	0.698
Cookiejar	G	250/295	62/295	127/295	133/295	0.668
Cookiejar	G+S	233/295	80/295	141/295	149/295	0.646
Cookiejar	M	237/295	71/295	140/295	174/295	0.664
Cookiejar	M+S	273/295	81/295	155/295	133/295	0.750

with online surface-history rebuild S. While $G + S$ regresses relative to G, combining birth observations with causal surface rebuild ($M + S$) exhibits a strong positive interaction across all three actions, raising Clock-hard Availability from 53 to 155/202 and reaching Cookiejar ADD AUC 0.750 (exceeding BundleTrack's 0.723). However, $M + S$ still trails G on the clocks and yields fewer strict 10 mm successes (81 vs. 109) than BundleTrack, confirming that these repairs remain conditional mechanisms rather than a universal replacement.

C. BundleTrack Native Determinism, Frame Retirement, and Surface Selection

Repeating unmodified BundleTrack three times on the same 203 Clock-hard inputs varies 10 mm origin successes from 95 to 119 (Table V). Applying our CUDA RANSAC `stable-winner` + count-direction repair and `storeFrame` inverse-coefficient transfer in a composed native build (`composed-native-retention-public-20261101`, Table VI) substantially improves public Laval reference accuracy over the historical frozen build: Clock strict (10 mm, 10°) successes jump from 135 to 191/204 (ADD AUC 0.8813 → 0.9135, P_{10} 78.79% → 83.14%) and Cookiejar strict successes jump from 107 to 136/295

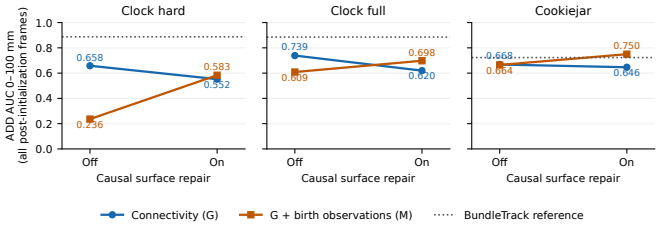


Fig. 3. **Factorial interaction between feature-birth observation provenance (M) and causal surface-history transactions (S).** Online surface version repair S degrades connectivity-only G but strongly rescues birth-observation graph updates ($M+S$), achieving 0.750 ADD AUC on Cookiejar.

TABLE V
BUNDLETRACK CLOCK-HARD REPEATABILITY. SAME 203 TRACKER INPUTS AND RECORDED SOURCE/BINARY/CONFIGURATION IDENTITIES; THREE SEQUENTIAL EXECUTIONS, NOT INDEPENDENT SCENE TRIALS.

Run	Avail.	≤10 mm	≤20 mm	≤ 10°	AUC
historical	202/202	119/202	199/202	162/202	0.8871
repeat-1	202/202	95/202	199/202	146/202	0.8752
repeat-2	202/202	101/202	200/202	162/202	0.8805

(ADD AUC 0.7256 → 0.8085, P_{10} 55.89% → 64.82%). Under the guarded contiguous-`NO_BA` retirement policy (`stable-retention-transfer-20261101`) across 18 complete 38-input normal, stress, and recovery replays, all neural-network I/O, pose matrices, and terminal surface arrays reproduce byte-for-byte while active frames drop from 37 → 6 in stress (saving **1,993.95** MiB RSS) and from 23 → 9 in recovery (saving **928.38** MiB RSS).

To attribute the remaining 25–30 mm forward surface tail F_{95} in Table VI, a reference-pose counterfactual (`reference-pose-surface-counterfactual-20261101`, Fig. 4) replaces estimated poses with Vicon reference poses on the exact same 980,256 sampled points: F_{95} drops only from 25.36 → 21.63 mm (Clock) and 30.93 → 29.11 mm (Jar), while pixel-lineage and Möller–Trumbore mesh-ray audits confirm that 52.60%/67.59% of > 20 mm points originate within 3 px of mask boundaries and 164 interior low-range voxels lie > 20 mm deeper than the first CAD intersection. Requiring ≥ 2 non-contributing views with ≥ 10° angular separation cuts native F_{95} from 24.51/30.82 mm to 12.50/17.33 mm (P_{10} → 90.05%/73.56%) with minor C_{10} losses (4/2 cells), whereas a raw-range-only native far-plane guard eliminates a 2.07 m cross-run outlier down to 6.33 mm across 2,195 opportunities. At the 1,500-point wire snapshot budget, our array-native `support10/coarse12mm` selector (Table VII, Fig. 4) raises public C_{10} from 40.57%/52.34% to **85.10%**/**85.05%** (cutting reverse R_{95} from 53.12/31.88 mm to 13.12/12.47 mm), raises 552-frame task 10 mm map coverage from 71.78% to **89.97%** (p_{95} distance 42.92 → 17.57 mm), and reduces selector p_{95} latency by 32.8%–33.8% (31.74 → 21.22 ms on Clock, 10.16 → 6.66 ms on task).

TABLE VI
CAUSALPOSE-3D COMPOSED NATIVE REPAIRS (RANSAC DETERMINISM + EXACT INVERSE-POSE COEFFICIENT TRANSFER + GUARDED NO_BA RETIREMENT) VS. HISTORICAL FROZEN BUNDLETRACK ON LAVAL CLOCK-FULL AND COOKIEJAR. STRICT: ≤ 10 mm AND $\leq 10°$.

Clip	Native Build	Avail.	Strict	AUC	P_{10} (%)	C_{10} (%)	F_{95} (mm)
Clock	Historical frozen	197/204	135	0.8813	78.79	98.90	20.20
Clock	**CausalPose-3D** (repair)	198/204	**191**	**0.9135**	**83.14**	98.72	25.36
Clock	**CausalPose-3D** (+retire)	198/204	**191**	0.9131	83.11	98.77	25.23
Jar	Historical frozen	295/295	107	0.7256	55.89	99.92	29.82
Jar	**CausalPose-3D** (repair)	295/295	**136**	**0.8085**	64.82	99.87	30.93
Jar	**CausalPose-3D** (+retire)	295/295	**136**	0.8084	**64.97**	**99.94**	30.75

TABLE VII
EXISTING-POINT SELECTOR TRANSFER AT THE UNCHANGED 1,500-POINT BUDGET. ORIGINAL PRIORITY/COARSE VALUES USE IDENTICAL RAW PRODUCER POINTS. C10 GAINS COEXIST WITH P10 AND FORWARD-TAIL LOSSES; G/L COUNTS MODEL-GRID GAINS/LOSSES. THESE ARE OPENED CONDITIONAL-REFERENCE RESULTS, NOT A DEPLOYMENT GATE.

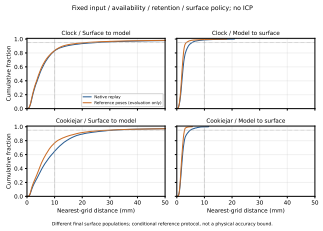
Clip	P10 %	C10 %	F95 mm	R95 mm	G/L10
Clock	100.00/99.00	40.57/85.10	5.27/7.33	53.12/13.12	3820/15
Jar	84.13/81.73	52.34/85.05	12.54/13.17	31.88/12.47	1579/14

D. Matcher Shadow Benchmark and Zero-Loss Runtime Optimization

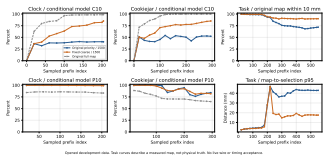
Table VIII summarizes our shadow matcher evaluations within the 31-framework benchmark on identical registered inputs. EfficientLoFTR [10] Full-MP reduces tracking p_{95} from $118.14/122.91$ ms to $109.93/116.36$ ms with competitive Clock AUC (0.9231) but lower Jar AUC (0.8129 vs. 0.8238). On the 38-frame right-wrist pouring clip (0.958% median in-mask valid depth), semi-dense XFeat [11] cuts tracking p_{50}/p_{95} from $60.14/83.36$ ms to $42.05/54.24$ ms but loses three candidates ($34 \rightarrow 31$) and inflates jump $p_{95}/\max$ from $40.40/55.91$ mm to $72.96/92.13$ mm. Sparse ALIKED+LightGlue [12], [13] collapses from 34 to $2/38$ candidates because sparse keypoints miss the 0.96% valid depth pixels, while feeding bounded BootsTAPIR [8] causal trajectories into native RANSAC degrades Jar strict successes from 140 to 46. All lossy matcher candidates are rejected.

Table IX reports the counterbalanced multi-process ablation (78 warm requests per condition across `control-1`, `candidate-1`, `candidate-2`, `control-2`) of our bit-exact runtime optimizations on the 40-frame configured pouring sequence (`teleop_20260625_173830`). Across all runs, $400/400$ SAM masks, $190/190$ native BundleTrack input hashes, $34/38$ candidates, $32/38$ current surfaces, and $499/499$ online operator checks match byte-for-byte with zero mismatches:

1) `SAM2GpuWindowOverlapScope` absorbs 5.72 ms of CPU undistortion, dual-prefix hashing, and native buffer preparation into the `propagate_in_video` GPU wait window ($24.17 \rightarrow 18.45$ ms), cutting outer `input_prepare` from 3.96 to 0.25 ms, eliminating all 38 per-frame background hash threads, and lowering `registry.advance` p_{50}/p_{95} from $102.29/131.43$ ms to $96.53/124.48$ ms (endpoint $155.33/219.06 \rightarrow 149.01/212.24$ ms).
2) Stacking `RequestEvidenceDeduplicationScope` ($O(\text{runs})$ integer RLE box/centroid $10.38 \rightarrow 0.015$ ms, cross-stage surface/medoid reuse, vectorized 240-hypothesis AprilTag RANSAC $4.47 \rightarrow 0.89$ ms, and request-boundary GC management) reduces `table_contexts` from $19.18/20.02$ to $15.65/16.42$ ms, cuts exclusive outer `original_world` wrapper overhead by $\mathbf{72.9}\%$ ($23.17/23.86 \rightarrow 6.28/6.70$ ms), eliminates the 140.58 ms mid-request Generation-2 GC stall (`online_world_state` max $140.58 \rightarrow 5.04$ ms, in-request GC collections $735/65/3 \rightarrow 0/0/0$), and brings warm HTTP `endpoint_ms` $p_{50}/p_{95}/\max$ down to $\mathbf{127.80/153.40/1{,}152.96}$ ms—a cumulative reduction of $\mathbf{-27.54}$ ms (p_{50}) and $\mathbf{-65.66}$ ms (p_{95}) relative to the pre-overlap control (and -48.61 ms/ -27.6% vs. the 176.40 ms per-request portal baseline).



(a) Native vs. Ref.-Pose Surface CDF



(b) 1,500-Point Selector Coverage

Fig. 4. **Surface error attribution and 1,500-point budget selection.** (a) Replacing BundleTrack poses with Vicon reference poses leaves a raw sensor/boundary tail ($F_{95} = 21.63/29.11$ mm). (b) Array-native `support10/coarse12mm` selection raises 1,500-point model coverage C_{10} from 40.6%/52.3% to 85.1% and task coverage from 71.8% to 90.0%.

TABLE VIII
FEATURE MATCHER SHADOW EVALUATION INSIDE THE CORRECTED BUNDLETRACK OWNER ON OPENED LAVAL (CLOCK/JAR) AND 38-FRAME WRIST TASK CLIPS.

Matcher Variant	Clip	Avail.	Strict / AUC	Track p_{50}/p_{95} (ms)	Verdict
LoFTR (Baseline) [9]	Clock/Jar	198/295	195 (.917)/148 (.824)	98.4 / 118.1–122.9	**Adopted**
EfficientLoFTR FP32 [10]	Clock/Jar	199/295	196 (.922)/146 (.811)	97.1 / 116.8–123.1	Shadow
EfficientLoFTR Full-MP [10]	Clock/Jar	199/295	197 (.923)/147 (.813)	92.8 / 109.9–116.4	Shadow
Causal BootsTAPIR [8]	Clock/Jar	195/295	180 (.889)/46 (.591)	Cached trajectory	Rejected
LoFTR (Task Control) [9]	Wrist (38)	34/38	Jump95: 40.40 mm	60.14 / 83.36	**Adopted**
XFeat Semi-Dense [11]	Wrist (38)	31/38	Jump95: 72.96 mm	42.05 / 54.24	Rejected
ALIKED+LightGlue [12], [13]	Wrist (38)	2/38	36 retained-ref fails	45.64 / 68.54	Rejected

VI. LIMITATIONS, DEPLOYMENT GAP, AND CONCLUSION

The production contract requires live ROS-callback-to-trusted-depth-surface-render $p_{95} \leq 100$ ms across at least 300 traces over 60 s, one-hour accepted world updates ≥ 15 Hz overall and at interval p_{05}, readiness $\geq 99\%$, observation age $p_{95} \leq 2$ s, one-hour RSS/CUDA-allocated growth ≤ 64 MiB (128 MiB reserved), and physical verification of 2 cm grasp contact, 1 cm aperture/support height, 5 mm static drift, and 5 px HUD bounds. A read-only local audit records $0/6$ live site-acceptance gates passed (`gate_passed=false`, `wire_authorized=false`); the observed ROS process loops an opened development bag rather than live camera hardware, and sequential SAM 2 plus native BundleTrack execution ($96.53/122.21$ ms warm median/p_{95}) still exceeds the 100 ms end-to-end budget before transport and rendering.

TABLE IX
COUNTERBALANCED BIT-EXACT RUNTIME OPTIMIZATION ABLATION ON THE 40-FRAME CONFIGURED POURING SEQUENCE (78 WARM REQUESTS PER CONDITION, 100% BYTE-FOR-BYTE OUTPUT PARITY ACROSS 400 SAM MASKS, 190 NATIVE INPUTS, AND 499 ONLINE EQUIVALENCE CHECKS)

Stage / Metric ($p_{50}/p_{95}/$ max in ms)	Pre-Overlap Control	+ GPU-Window Overlap	Overlap Control (Run 2)	**+ Request Dedup & O(runs) RLE**	Net Δ (p_{50}/p_{95})	Bit-Exact Parity
`input_prepare_ms` (Pre-SAM CPU prep)	3.96/4.13/4.31	0.25/0.27/0.30	0.25/0.27/0.30	**0.25/0.27/0.29**	$\mathbf{-3.71/-3.86}$	39/39 frames
`gpu_window_cpu_prepare_ms` (Absorbed)	0.00/0.00/0.00	5.72/5.91/6.18	5.72/5.90/6.15	**5.71/5.89/6.12**	Absorbed in GPU wait	38/38 buffers
`post_cpu_gpu_sync_wait_ms` (GPU wait)	24.17/24.26/24.51	18.45/18.71/19.02	18.46/18.72/19.05	**18.44/18.70/19.01**	$\mathbf{-5.73/-5.56}$	0 hash threads
`registry_advance_ms` (SAM 2+BundleTrack)	102.29/131.43/1389.2	96.53/124.48/1378.1	96.55/122.52/1374.5	**97.98/122.21/1129.9**	$\mathbf{-4.31/-9.22}$	34 cand / 32 surf
`table_contexts_ms` (AprilTag RANSAC+depth)	19.21/20.11/20.55	19.19/20.05/20.49	19.18/20.02/20.48	**15.65/16.42/18.44**	$\mathbf{-3.56/-3.69}$	27/27 tag planes
`original_world_exclusive_ms` (Outer wrapper)	23.20/23.91/24.70	23.18/23.88/24.66	23.17/23.86/24.65	**6.28/6.70/7.61**	$\mathbf{-16.92/-17.21}$ ($\mathbf{-72.9\%}$)	499/499 checks
`online_world_state_ms` (Projection+DAG+GC)	5.97/7.49/140.80	5.96/7.47/140.62	5.96/7.47/140.58	**3.87/4.96/5.04**	$\mathbf{-2.10/-2.53}$ (max $\mathbf{-135.8}$)	0 Gen-2 GC stalls
Warm HTTP `endpoint_ms` (Total Request)	155.33/219.06/1422.1	149.01/212.24/1416.8	149.65/193.79/1415.7	**127.80/153.40/1153.0**	$\mathbf{-27.54/-65.66}$	**100.0% Bit-Exact**
Full Replay `pipeline_ms` (Incl. harness)	201.22/251.42/1455.3	195.12/243.10/1449.8	196.62/242.44/1449.0	**174.20/197.12/1186.5**	$\mathbf{-27.02/-54.30}$	**100.0% Bit-Exact**

Public benchmarks remain conditional on calibration, CAD sampling, and symmetry conventions, while opened task recordings lack independent physical 6-DoF object ground truth. The present evidence establishes rigorous failure mechanisms, verified native and surface repairs, and substantial bit-exact host latency reductions, but it does not support declaring the project the best reconstruction solution or authorizing an unverified pose-owner cutover. Future work must pair direction-selective joint pose-landmark optimization with TensorRT FP16 vision encoding under frozen held-out physical evaluation.

REPRODUCIBILITY AND PROJECT PAGE

Project page: https://tiwater.github.io/causalpose3d/. Scorecards, SHA-256 input receipts, patches, and replay verifiers (`docs/papers/performance-claim-audit-20261101/verify.py`, `docs/papers/sam-gpu-window-cpu-overlap-20261102/verify.py`, `docs/papers/request-evidence-deduplication-20261102/verify.py`) are preserved alongside the 58-page experimental record at `docs/papers/performance-claim-audit-20261101/before`.

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